ModelIt!

Student Activity Pack

Quantum Computing: Why 0 and 1 Aren't Enough

Modeling the Physics and Engineering of Computation Beyond Classical Limits

Grade 10L3 | NGSS HS-PS4-3, HS-PS2-5 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is the difference between a classical bit (0 or 1) and a quantum bit (qubit)?

2 Why do you think quantum computers need to be cooled to near absolute zero?

3 Draw what you think happens when a qubit is in superposition versus when it is measured.

4 If quantum computers are so powerful, why don't we use them for everything already?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Qubit Coherence Time | Entanglement Fidelity | Error Rate | Gate Operation Speed | Cooling Temperature | Quantum Volume | Decoherence Rate | Classical Overhead | Computational Advantage

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current Technology

2024-era qubit parameters with 100 microsecond coherence

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Near-Perfect Qubits

Error rate reduced to 0.01%, coherence extended to 10 ms

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Thermal Disruption

Cooling temperature rises from 15 mK to 100 mK

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Qubit Connectivity — The pattern of which qubits can directly interact with which others — full connectivity allows any two qubits to entangle directly, while limited connectivity requires extra operations to route information

Readout Fidelity — The accuracy with which the final quantum state can be measured and converted to a classical result — imperfect readout adds another layer of error to quantum computation

Crosstalk Interference — Unwanted coupling between neighboring qubits that causes operations on one qubit to disturb adjacent qubits — a major source of correlated errors in dense qubit arrays

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why would a computer that uses the weirdest rules in physics be millions of times faster than the most powerful supercomputer ever built? Use evidence from your simulation to support your answers.

1 How does your model demonstrate that quantum computing is fundamentally a race against decoherence?

2 Why does the error correction overhead create a paradox — and how might future technology resolve it?

3 What does your model predict about the minimum qubit quality needed for quantum advantage on real problems?

4 How would your model change if a room-temperature qubit technology were invented?

5 What are the limitations of modeling quantum computation with classical simulation tools?

8 STEM Challenge: Design a Quantum Error Correction Strategy

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YOUR ENGINEERING MISSION

Design an error correction approach for a 1,000-qubit quantum processor that maximizes the number of usable logical qubits while keeping total error rates below the threshold for useful computation.

THE SCENARIO

A quantum computing company has built a 1,000 physical qubit processor with a per-gate error rate of 0.3% and a coherence time of 200 microseconds. They need your team to design an error correction strategy that determines how many physical qubits to dedicate to error correction versus computation, and whether the resulting logical qubits can solve problems that justify the system's $50 million cost.

Engineering Design Thinking

1. WHAT ERROR CORRECTION CODE WILL YOU USE AND WHAT IS ITS QUBIT OVERHEAD RATIO?

2. HOW MANY LOGICAL QUBITS DOES YOUR STRATEGY PRODUCE FROM 1,000 PHYSICAL QUBITS?

3. WHAT CIRCUIT DEPTH CAN YOUR LOGICAL QUBITS SUPPORT BEFORE ACCUMULATED ERRORS EXCEED YOUR CORRECTION CAPACITY?

4. WHAT SPECIFIC REAL-WORLD PROBLEM COULD YOUR SYSTEM SOLVE THAT A CLASSICAL COMPUTER CANNOT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Quantum Computing Engineers design and optimize quantum processors, error correction schemes, and quantum algorithms. They work at IBM, Google, Microsoft, Amazon, and startups like IonQ, Rigetti, and PsiQuantum, earning $120,000-$250,000/year. This is one of the fastest-growing and highest-paying fields in technology.